# EGYPT'S REPORT
# ON THE IMPLEMENTATION OF THE
# 2021 RECOMMENDATION ON OPEN SCIENCE

## QUESTIONNAIRE

## A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: **Egypt**
- Organization(s) or entity(s) responsible for the preparation of the report*:

  Name: **Ministry of Higher Education & Scientific Research**

  Website: **https://mohesr.gov.eg**

  Email address*:

  Phone number*:

  Please describe the role/mandate of your organization:
- Officially designated contact person(s) who completed this survey:

  Full name*: **Majid M. Al-Sadek**

  Position*: **Secretary-General, Egyptian Knowledge Bank, and Acting Director of Egyptian National STI Network.**

  Email address*:
- Full Name(s) of designated official(s) certifying the report:
- Brief description of the consultation process established for the preparation of the report:
- Other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey:

  Name of the organization*: Egyptian Science Technology and Innovation Observatory – Acadey of Scientific Research & Technology & Supreme Council of Universities – Ministry of Higher Education & Scientific Research

  Website*: **http://www.asrt.sci.eg**

  Email address*:

  Sector: Public

## B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1.1 Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

**Yes,** Egypt is actively promoting Open Science. The Egyptian Knowledge Bank (EKB) is a significant initiative that provides access to a vast array of resources. Awareness is growing, particularly within research, education, and policy circles. The National STI Strategy emphasizes knowledge dissemination, which aligns with Open Science principles. Several ministries and national authorities in Egypt have been actively involved in promoting the 2021 UNESCO Recommendation on Open Science. These include:

- **Ministry of Higher Education and Scientific Research:** The ministry is central to Egypt's science and research policy and plays a crucial role in promoting Open Science principles within higher education & research institutions. The STI strategy document also highlights the role of MoHESR in achieving national research and development goals and promoting the key principles of Open Science.
- **The Academy of Scientific Research and Technology (ASRT):** ASRT is a pivotal national organization entrusted with the responsibility of planning, promoting, and coordinating scientific research and technological development

within Egypt. It actively engages in international collaborations in the scientific and technological domains, including the promotion of Open Science practices. ASRT has consistently provided support and funding for a national project aimed at developing, publishing, and transitioning Egyptian periodicals to an Open Access Model through partnerships with esteemed publishing houses such as Elsevier, Taylor & Francis, Wolters Kluwer, and Digital Commons. This initiative has facilitated the publication of 132,244 articles across 86 supported journals, thereby enabling their inclusion in international abstract and citation indexes. enabling their inclusion in international abstract and citation indexes.

- **The Science & Technology Development Fund (STDF):** in Egypt plays a crucial role in advancing scientific research and technological innovation within the country. STDF supports the promotion of open science, through funding the Transformative Agreements: a landmark agreement with Springer Nature, provide Egyptian researchers with opportunities to publish in leading international journals while ensuring open access.
- **The Egyptian Knowledge Bank (EKB):** EKB is a significant national initiative that provides access to a vast collection of scientific resources. While its primary goal has been to democratize access to knowledge, it inherently aligns with Open Science principles. The EKB's efforts contribute to making research and data more accessible. EKB, additionally promotes open science on the national level, through the series of running workshops, info-days, and professional development courses. EKB in collaboration with the Academy of Scientific Research and Technology (ASRT), has embarked on a project to create and operate a national-scale open access publishing platform. This platform houses over 1045 journals and more than 390,000 articles, providing a robust infrastructure for the global dissemination of scientific research.

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

**Yes,** Arabic Language.

1.3 Have there been awareness raising activities on open science, including all its key elements, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities?

**Yes**, through national initiatives such as the Egyptian Knowledge Bank (EKB), Egypt organizes workshops and awareness activities that promote the value and principles of Open Science. The STI strategy's emphasis on international cooperation underscores Egypt's commitment to fostering open dialogue with other knowledge systems. Collaborations with international partners and participation in global scientific initiatives facilitate knowledge exchange and contribute to the broader promotion of open science. Initiatives like the EKB and the Egyptian Innovation Bank (EIB), which foster an environment that supports innovation, including open innovation, indirectly foster public engagement and raise awareness in Open Science by providing broader access to knowledge.

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science, in publicly funded research?

**Yes,** Egypt has been actively undertaking and planning specific actions to incorporate the values and principles of Open Science into publicly funded research. These efforts are in line with the 2021 UNESCO Recommendation on Open Science. Here are some key actions and plans:

**National STI Strategy:**

- Egypt's National STI Strategy provides a framework that emphasizes knowledge sharing, international collaboration, and transparency in research, which are core tenets of Open Science.
- The strategy directs publicly funded research to adhere to principles that promote accessibility, inclusivity, and responsible conduct of research.
- This strategic document guides funding policies and prioritizes projects that align with Open Science values.

**Funding Policies and Grant Requirements:**

- Public funding bodies, such as the Academy of Scientific Research & Technology (ASRT), Science & Technology Development Fund (STDF), are increasingly incorporating Open Science requirements into grant guidelines.
- Funded research projects may be mandated to share data, publish findings in open access journals, or use open-source software and tools.
- These requirements aim to ensure that publicly funded research is accessible, reusable, and contributes to the global knowledge commons.

**Investment in Open Science Infrastructure:**

- Egypt is investing in open science infrastructure, such as the Egyptian Knowledge Bank (EKB), national open access publishing platform, the national project for developing and transitioning Egyptian journals to open access model, transformative open access agreement, National LMS platform, and the Egyptian Knowledge Graph (EKG), Funding of Arabic Citation Index**,** National Bank for Scientific Laboratories and Equipment, which are supported by public funding.
- These infrastructures facilitate the sharing of research outputs and data transparently and inclusively monitor and assess research performance,, promoting Open Science principles.
- Continued investment in and development of these infrastructures will further solidify the integration of Open Science in publicly funded research.

**Capacity Building Programs:**

- Publicly funded institutions and organizations conduct capacity-building programs for researchers on Open Science practices.
- These programs train researchers on data management, open access publishing, and the use of open-source tools.
- By enhancing the skills and knowledge of researchers, these programs promote the adoption of Open Science practices in funded projects.

**Monitoring and Evaluation:**

- Efforts are underway to monitor and evaluate the implementation of Open Science policies in publicly funded research.
- Indicators and metrics are being developed to track progress and identify areas for improvement.
- This monitoring will ensure that funded research adheres to Open Science principles and contributes to broader scientific goals.

**Open Access Funding Policies:**

- There is a growing emphasis on open access publishing for research funded by public institutions.
- Policies and incentives are being considered to encourage or mandate open access publication of research findings.
- These policies aim to maximize the impact of publicly funded research by making it freely available to the widest possible audience.

Through these actions and plans, Egypt is actively working to embed Open Science values and principles into its publicly funded research system. This commitment ensures that research investments translate into broader scientific progress and societal benefits.

# 2. Developing an enabling policy environment for open science

### 2.1 Does your country have a national policy, strategy or plan of action on science, technology and innovation (STI)?

**Yes**, Egypt has a national policy on Science, Technology, and Innovation (STI), known as the National Strategy for Science, Technology, and Innovation 2030 (NSSTI 2030). This policy was developed by the Ministry of Higher Education and Scientific Research (MoHESR) and was first adopted in 2016, with its most recent update in 2019. The development process involved universities, research institutions, the Academy of Scientific Research and Technology (ASRT), government agencies, and private sector representatives, ensuring a comprehensive approach to advancing STI in Egypt.

The strategy outlines key components, including policy and legislation for science and technology, human resource and infrastructure development, international leadership in research, investment in scientific research, and integration of STI with industry and education. Additionally, it focuses on strategic sectors such as energy, water, health, agriculture, ICT, and environmental sustainability. The policy indirectly promotes Open Science principles and values. Although it may not always explicitly use the term "Open Science," it emphasizes principles such as transparency, collaboration, and knowledge sharing. Furthermore, it incorporates open science principles to promote research accessibility and collaboration.

A crucial aspect of Egypt's STI policy is its commitment to open science principles, which promote transparency, accessibility, and collaboration in research. The policy supports open access to scientific knowledge, exemplified by initiatives like the Egyptian Knowledge Bank (EKB), which provides free access to research databases for Egyptian students and researchers. Furthermore, Egypt has been actively investing in open science infrastructure, including digital platforms, cloud computing, and high-performance computing systems to support scientific collaboration and data sharing.

In addition to open access and infrastructure, Egypt's STI policy emphasizes the engagement of societal actors in scientific activities. There is a strong focus on linking universities, private companies, and civil society organizations to foster innovation and knowledge exchange. Programs and funding mechanisms are in place to encourage researchers and innovators to develop solutions that address real-world challenges. The policy also promotes open dialogue with other knowledge systems, facilitated through institutions such as ASRT, which acts as a central hub for coordinating scientific collaborations at the national and international levels.

### 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s) at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science?

Egypt does not have a wide national policy, or laws specifically dedicated to open science fully in accordance with the 2021 UNESCO Recommendation on Open Science. However, several initiatives and agreements have been established that partially address core elements of open science, particularly transparency, collaboration, and knowledge sharing, through active support for open access to knowledge and research.

One of the most important initiatives is the **Egyptian Knowledge Bank** (EKB), which was launched in 2016. The EKB is a national online digital library that provides free access to a vast pool of scientific publications, journals, and educational material for all Egyptian citizens. The initiative is of critical importance in promoting open access to scientific literature, which is the core aspect of open science.

In addition to the EKB, Egypt has also signed a transformative agreement with leading academic publishers to facilitate open access publishing for Egyptian researchers. Furthermore, Egypt launched a national project for developing and transitioning Egyptian journals to an open access model with leading international publishers. Additionally, Egypt developed a national open access publishing platform. For instance, in December 2021, Egypt signed its first transformative agreement with Springer Nature, enabling researchers to publish open access articles without incurring article processing charges. Similarly, a previous agreement with IOP Publishing allowed Egyptian researchers to publish open access articles, although the agreement expired at the end of 2022.

Another achievement is the establishment of the Open Science Community Egypt (OSCE) in September 2023. OSCE is Egypt's inaugural open science network, founded by professionals in librarianship and archiving careers. The community is in place to set the bar higher in open science practice awareness and enhance Egypt's research reputation worldwide through the adoption of open science policies and frameworks.

### 2.3 In your country, are there specific policy instruments that aim to promote open science?

Egypt has specific policy instruments that aim to promote open science, support access to scientific knowledge, and enhance research transparency. Various initiatives and agreements have been implemented to align with the key elements of open science as mentioned above, ensuring that research results and knowledge resources are widely accessible. One of the most important initiatives is the Egyptian Knowledge Bank (EKB), which provides free access to a wide range of scientific publications, journals, and educational materials for all Egyptian citizens. This initiative plays a crucial role in democratizing access to knowledge and supporting researchers and students in various disciplines. Through the EKB, Egypt promotes open access to scientific literature, capacity building for researchers, and digital infrastructure for research dissemination.

Another key player in promoting open science is the Academy of Scientific Research and Technology (ASRT). ASRT plays a pivotal role in developing and transitioning Egyptian journals to an open access model with leading international publishers, thereby elevating the level of local journals and establishing international presence and dissemination of scientific knowledge. Furthermore, ASRT collaborates with the Egyptian Knowledge Bank (EKB) to host and operate the Egyptian open access publishing platform, which houses over 1040 journals. Additionally, ASRT makes patents publicly available by publishing them, ensuring that innovations can be utilized and built upon by researchers, entrepreneurs, and the general-public, thereby fostering collaboration between academia and industry. In addition, the Academy of Scientific Research and Technology publishes books for researchers, making critical academic knowledge widely accessible. These efforts promote knowledge dissemination, encourage innovation, and support researchers by providing free access to high-quality academic materials.

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation?

Under development.

In conjunction with Egypt's progress in democratizing access to scientific, educational, and general knowledge through various national-scale initiatives, such as the Egyptian Knowledge Bank (EKB), the national project for elevating and transitioning Egyptian journals to an open access model through agreements with international leading publishers, the transformative open access agreement, the National Open Access Publishing Platform, the Egyptian Innovation Bank, the S&T Grid & Cloud Computing Centre, and the National Learning Management System (LMS) digital platform, Egypt is developing a national strategy for open science, led by the Ministry of Higher Education and Scientific Research. This strategy aims to establish legal and institutional frameworks that promote open access to research, encourage data sharing, and integrate open science principles into national research policies. The objective is to ensure that scientific knowledge is widely accessible to researchers, institutions, and the public, thereby enhancing the impact of research within the country. In support of Recommendation 17.b, Egypt is actively studying and developing legal and regulatory frameworks to institutionalize open science practices.

In response to Recommendation 17.c, Egypt has already implemented funding mechanisms and sustainable initiatives to support open science. The Egyptian Knowledge Bank provides free access to scientific literature, educational resources, and research tools, making it one of the largest national knowledge repositories. Furthermore, Egypt has signed transformative agreements with major academic publishers, such as Springer Nature, enabling researchers to publish open-access articles without incurring article processing fees. The Egypt Open Access Publishing Platform has been developed and launched, upskilling Egyptian journals and transitioning them to an open access model with subsidies for publishing agreements with international publishers. Additionally, Egypt has developed and expanded the National Research & Education Network and is funding international connectivity to global research and education networks. These funding efforts ensure that open science initiatives remain financially viable and accessible to a broad range of researchers. To ensure coordination among national stakeholders, as outlined in Recommendation 17.f, several governmental and institutional entities are actively involved in the open science movement in Egypt. Key organizations include the Ministry of Higher Education and Scientific Research, the Academy of Scientific Research and Technology, the Supreme Council of Universities, and the Egyptian Knowledge Bank.

Egypt actively engages in international open science collaboration, aligning its national strategies with global standards as outlined in UNESCO's Recommendation 17.h. This commitment is demonstrated through close partnerships with UNESCO and other international organizations. Furthermore, Egypt fosters international knowledge exchange through participation in global initiatives, collaborations with academic publishers, and strategic partnerships designed to elevate the international prominence of Egyptian research. Notably, regional agreements with the Association of Arab Universities and the Federation of Arab Scientific Research Councils have been established to expand the reach of the Egyptian Knowledge Bank International (EKB Int.) and support the development of a regional open science platform. These concerted efforts solidify Egypt's role as a significant contributor to the global open science movement.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies?

While overarching national strategies and supporting digital platforms and tools, such as the Egyptian Knowledge Bank (EKB), the national open access publishing platform, transformative agreements, the Egyptian Innovation Bank, and others, provide an enabling framework, the implementation of open science principles also relies on individual research institutions, universities, and scientific societies. The Ministry of Higher Education and Scientific Research is actively involved in the process of implementing the national STI strategy and, therefore, exerts influence on universities and research institutions within Egypt to encourage wider institutional adoption. Key national actors that actively promote open science at the institutional level include:

- The Academy of Scientific Research and Technology (ASRT)
- The Science and Technology Development Fund (STDF)
- The Supreme Council of Universities
- The Council of Research Centers and Institutes (CRCI)

ASRT funds research projects and hosts 20 sectoral councils in all scientific fields, with a specific focus on the medical research, ethics, and drug development fields. There is a growing awareness of the importance of open access publishing, data sharing, and collaborative research practices within Egyptian institutions.

Egypt At the governmental level, Egypt has participated in bilateral scientific agreements such as the EU-Egypt Scientific and Technological Cooperation Agreement, signed in 2005 and implemented in 2008. This agreement focuses on critical areas such as food security and the water-food-energy nexus, and supports international scientific cooperation as recommended by UNESCO. The Academy of Scientific Research and Technology (ASRT) and the Science and Technology Development Fund (STDF) in Egypt also fund research projects and host sectoral councils in the field of medical research, ethics and drug development. This initiative supports UNESCO recommendations on promoting a strong scientific culture and encouraging innovative approaches to research.

A key civil society initiative in Egypt is the Open Science Community of Egypt (OSCE), founded in September 2023 by three Egyptian women with backgrounds in librarianship and archiving. The initiative aims to raise awareness of open science, improve Egypt’s global ranking in research, and address misconceptions about open science within the academic community. It directly supports UNESCO’s recommendations on transforming scientific culture and promoting innovative approaches.

The UNESCO Arab Science Platform, developed by the UNESCO Office in Cairo, is another important initiative that promotes scientific exchange and cooperation within the Arab region. By facilitating discussions among researchers, this programme fosters international scientific partnerships and addresses the recommendation to enhance international cooperation.

2.6 In your country, are there specific funding mechanisms for open science at the national level?

**Yes,** Egypt is progressively integrating Open Science into its research funding policies. The Academy of Scientific Research & Technology (ASRT) and the Science and Technology Development Fund (STDF) provide research grants that indirectly support Open Science by funding collaborative projects, open-access publications, and international research partnerships. These grants align with UNESCO's call for funding policies that support open research, open data, and inclusive participation.

Egypt is actively investing in open-access digital infrastructures, exemplified by the Egyptian Knowledge Bank (EKB), Egyptian Knowledge Graph, Egyptian Innovation Bank, the national open access publishing platform, transformative open access agreements, supporting Egyptian journals through international open access publishing, connectivity to global research & education network, funding Science and Technology Grid and Cloud Computing Centre (STGC[3]), and the national Learning management system. This initiative aligns with UNESCO's recommendation to develop sustainable platforms that facilitate knowledge exchange between researchers and society.

2.7 In your country, have open science practices been integrated in existing research funding mechanisms?

**Yes,** In Egypt, open science practices have been integrated into existing research funding mechanisms, particularly through the efforts of the Academy of Scientific Research and Technology (ASRT) and the Science and Technology Funding Authority (STDF) in collaboration with the Egyptian Knowledge Bank (EKB). For instance, ASRT funding has supported over 15 years several agreements with international publishers to elevate and transition Egyptian journals to an open access model.

ASRT, in collaboration with EKB, has funded the hosting and operational services for the national open access publishing platform, which houses 1045 journals and is globally accessible. This platform promotes the principles of open science by making research outputs freely available to the public. Furthermore, ASRT launched the Egyptian Innovation Bank with funding to specific projects that address global and national challenges and serve as a trusted matchmaking and marketplace for innovative ideas and solutions, fostering the convergence between academia, open innovation, and industry. Additionally, ASRT funding and co-develop the Egyptian Knowledge Graph, which fosters an inclusive, equitable, and holistic open valuation and analytics framework for the Egyptian research landscape.

STDF, in collaboration with EKB, has facilitated a transformative agreement with Springer Nature that enables Egyptian researchers from over 150 institutions to publish their work open access without Article Processing Charges (APCs). This agreement covers both fully open access and hybrid journals.

These initiatives support the integration of open science practices within funding mechanisms by covering open access publication fees.
Egypt's involvement in international research programs such as Horizon Europe and partnerships with entities like UNESCO for open science in the Arab region demonstrate a broader integration of open science principles. While not directly a funding mechanism, these engagements influence how research funding is approached, emphasizing openness.

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation?

**Yes,** Egypt is actively fostering and will continue to foster equitable public-private partnerships on Open Science through various initiatives and strategies. These efforts align with the 2021 UNESCO Recommendation on Open Science. Following some highlights:

***Legislative and Policy Frameworks***
**Law No. 23 of 2018 Promulgating the Law on Incentives for Science, Technology, and Innovation:**

The legal framework facilitates public-private partnerships by providing novel avenues for higher education and research institutions to establish companies independently or in collaboration with private entities or individuals. This enables the commercialization and exploitation of research project outcomes. Furthermore, the legislation offers tax incentives to companies and individuals who contribute to research projects. These incentives can be utilized to establish startups, spin-offs through science and technology parks, and technology incubators in partnership with private entities. The low barriers to entry in this area are as follows:

- **Establishment of science and technology valleys:** These are areas with technology incubators and companies that promote innovation, technology development, transfer, and commercialization.
- **Establishment of technology incubators:** These provide business services, technical and scientific facilities for research projects, support mechanisms, and technical consultations for innovators and start-up companies.
- **Commercial exploitation of research:** Higher education and research institutions can utilize scientific research to advance society and generate funding in collaboation with private companies.
- **Funding of research projects:** Companies and individuals can fund research projects, and this funding is considered a deductible expense for tax purposes.. These tax incentives encourage private sector involvement in scientific research and development.

***Financial Mechanisms and Initiatives***

**Egyptian Innovation Bank (EIB):**

The Egyptian Innovation Bank (EIB) plays a crucial role in encouraging public-private partnerships. The EIB supports initiatives that promote collaboration between the public sector, private companies, academia, and industry. By creating an environment that values and supports these partnerships, the EIB helps to advance Open Science practices and innovation in Egypt. One gateway for investors, innovators and service providers (supported by matchmaking tools), EIB Networks screen qualified solutions for more investments/support. EIB features:

- **Comprehensive searchable database of innovations**.
- **Comprehensive database of innovation enablers in Egypt** (Venture capitalists, funding agencies, co-working spaces, patent attorneys, business plan competitions, etc)
- **Interactive map of incubators** in Egypt (~140 incubators)
- **Marketplace** that facilitates transactions at multilevel cooperation (including innovators-innovators and innovators-investors collaboration)
- **Structured Support services including** IPR-helpdesk, legal advice, funding, technology transfer and administrative support
- **Tech Portfolio**: interactive catalogue of innovative products resulted from **industry-academia collaboration.**

**Knowledge Technology Alliance (KTA):**

The Academy of Scientific Research and Technology (ASRT) provides programs such as the Knowledge Technology Alliance (KTA) which supports public-private partnerships in open science. KTA projects directly facilitate collaboration between research institutions and industry partners, promoting knowledge exchange and the development of innovative solutions. These alliances are essential for advancing Open Science goals and ensuring that research outcomes are translated into practical applications.

**Alliance and Development initiative:** Launched by Egypt's Ministry of Higher Education and Scientific Research (MOHESR), as a presidential program designed to enhance the impact of scientific research on societal advancement. With a substantial investment of EGP 1 billion, this initiative aims to form strategic alliances that stimulate innovation and entrepreneurship by fostering partnerships between public and private sectors. A key objective is to maximize the outputs of scientific research, aligning them with the Egyptian government's 2024-2026 work program to address national development priorities. By promoting collaborative efforts, the initiative seeks to bridge the gap between academia and industry, ensuring that research findings translate into practical applications that benefit society at large. This approach not only accelerates technological progress but also embodies the principles of open science, encouraging transparency, accessibility, and the democratization of knowledge.

***Research and Innovation Initiatives and Bodies***
**Generative Innovation Index:**
Egypt's Generative Innovation Index is another important tool in promoting public-private partnerships. This index measures innovation performance across various Egyptian ecosystems, which inherently involves assessing and encouraging collaboration between different sectors. By highlighting successful partnerships and identifying areas for improvement, the index supports the growth of public-private collaborations in Open Science.

Governmental entities such as the **Science & Technology Development Fund (STDF)** are positioned to facilitate such partnerships. Their mandate to support innovation and technology transfer aligns with the goals of open science.

**The Establishment of the Innovators Support Fund (Egypt):** an Egyptian fund that aims to discover, train, support, and fund gifted individuals, entrepreneurs, and innovators. It was established by Law No. (1) of 2019 and reports to the Minister of Higher Education and Scientific Research. Will further strenghthen Egypt to encourage equitable and more balanced public-private partnership on Open Science.

**City of Scientific Research and Technological Applications (SRTA-City) in Egypt and Its Role in Open Science:** Established in 1993 by Presidential Decree No. 85 and officially inaugurated in 2000, is one of Egypt's leading research hubs. Located in **Borg El Arab, Alexandria**, and spanning **225 acres**, SRTA-City operates under the Ministry of Higher Education and Scientific Research. Its primary mission is to advance scientific research in key areas aligned with national development goals, including **biotechnology, information technology, advanced materials, environmental sciences, and agricultural research**. The city hosts several research institutes, such as the **Genetic Engineering and Biotechnology Research Institute** and the **Institute of Advanced Technologies and New Materials**, fostering interdisciplinary collaboration. SRTA-City actively supports **open science principles** by promoting **knowledge sharing, interdisciplinary research, and collaboration with universities, research centers, and industries** at both national and international levels. Through **open-access research projects, public-private partnerships, and international scientific cooperation**, the city contributes to making scientific discoveries more accessible. Additionally, SRTA-City's role in hosting **scientific conferences, innovation hubs, and incubators** helps in democratizing knowledge, ensuring that research findings benefit society, industry, and policymaking. By integrating open science practices, SRTA-City plays a crucial role in **bridging the gap between research and application**, aligning with Egypt's vision for sustainable development and global scientific collaboration.

**The Information Technology Industry Development Agency (ITIDA)**: in Egypt plays a pivotal role in fostering public-private partnerships, particularly within the technology sector. ITIDA actively engages in connecting industry stakeholders with academic and research institutions, which is a cornerstone of fostering innovation and knowledge sharing. ITIDA's initiatives prioritize supporting startups and entrepreneurs, often involving collaborations with both public and private entities. Through the ITAC program, ITIDA promotes collaboration between ICT companies and Egyptian researchers.

Through these initiatives and bodies, Egypt is making substantial efforts to create a collaborative ecosystem that supports Open Science and leverages the strengths of both the public and private sectors. This approach ensures that Open Science practices are not only developed but also effectively implemented and utilized for societal benefit.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy.

**Yes,** Egypt is actively implementing and developing a national monitoring framework for Open Science, integrated within its broader Science, Technology, and Innovation (STI) policy. Following are key components and efforts:

**Egypt STI Strategy:**

The Egypt National STI Strategy document outlines the country's commitment to improving its scientific research and innovation ecosystem. It emphasizes the importance of monitoring and evaluating the effectiveness of STI policies, which includes aspects related to Open Science. The strategy aims to:

- **Enhance Data Collection and Analysis:** The strategy emphasizes the need for robust data collection and analysis mechanisms to track progress in STI, including indicators related to research output, collaboration, and knowledge sharing, all of which are relevant to Open Science.
- **Improve Evaluation Processes:**There is a focus on developing evaluation processes that align with international standards, allowing for comparisons and benchmarking. This includes evaluating the impact of STI policies on research accessibility and collaboration.
- **Promote Transparency and Accountability:** The STI strategy promotes transparency in research and development activities, which aligns with Open Science principles. Monitoring mechanisms help ensure accountability and effectiveness of policies.

**Egyptian Knowledge Graph (EKG):**

The Egyptian Knowledge Graph (EKG), developed by the Academy of Scientific Research & Technology in collaboration with Elsevier and Scimago Labs is a significant initiative that contributes to the monitoring and analysis of research performance. The EKG is a comprehensive big-data research performance analytics platform. **A key strength of the EKG is its inclusivity:** it evaluates the *entire* Egyptian research output, encompassing *all* subject areas, including locally published and Scopus-indexed articles, *regardless of language*. This ensures language equity in research evaluation, giving previously unmeasured Arabic social sciences and humanities publications the recognition they deserve. As a research analytics & insights big data platform, EKG:

- **Collects and Integrates Data:** EKG gathers data from various sources, including research publications, preprints, patents, and institutional information. This integration allows for comprehensive monitoring of research activities and trends.
- **Provides Analytical Capabilities:** EKG offers analytics tools to assess research output, identify collaboration patterns, and track the impact of research. These analytical capabilities are essential for monitoring Open Science indicators, such as the accessibility and reuse of research data.
- **Supports Evidence-Based Policymaking:** By providing reliable data and insights, EKG supports evidence-based policymaking in STI. This includes monitoring the effectiveness of Open Science policies and initiatives. Creating a more holistic researcher profiles and organization profiles.
- **Evaluate the effectiveness of Open Science policies:** Monitoring collaboration metrics allows us to gauge the success of policies aimed at promoting open research practices.
- **Monitor inclusivity:** measuring collaboration can help show if research is being conducted in equitable ways, and if researchers from diverse backgrounds are able to participate.
- **The EKG indexes:** 1,904,026 articles, 1,034 Egyptian journals, and includes over 522,000 author profiles and 170 research organization profiles.

**Egyptian STI Observatory (ESTIO):**

The Egyptian STI Observatory (ESTIO) plays a crucial role in monitoring and reporting on Open Science implementation within the broader STI landscape. ESTIO's responsibilities include:

- Monitoring STI Indicators: ESTIO monitors key STI indicators, including those related to research and development, innovation, and knowledge sharing. This includes indicators that reflect Open Science practices, such as open access publishing and data sharing.

- Reporting and Analysis: ESTIO provides reports and analyses on STI trends and developments, informing policymakers and stakeholders about progress and challenges. These reports can include assessments of Open Science implementation and its impact on the research ecosystem.
- Supporting Policy Evaluation:ESTIO's work supports the evaluation of STI policies, including those related to Open Science. By providing data and analysis, ESTIO helps assess the effectiveness of these policies and identify areas for improvement.

In summary, Egypt is in the actively developing and strengthening its national monitoring framework for Open Science. The STI Strategy provides the policy direction, the Egyptian Knowledge Graph offers the technological infrastructure for data collection and analysis, and the Egyptian STI Observatory plays a key role in monitoring, reporting, and evaluating Open Science implementation. These combined efforts contribute to a comprehensive approach to monitoring and promoting Open Science in Egypt.

# 3. Investing in open science infrastructures and services

3.1 What is the latest official data for your country’s percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers.

As of 2023, Egypt’s research and development (R&D) expenditure was reported as 1.02% of its Gross Domestic Product (GDP). This data was published by the United Nations Educational, Scientific and Cultural Organization (UNESCO) and the World Bank, which underscores Egypt’s commitment to investing in R&D.

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers.

As of December 2024, Egypt’s internet penetration rate has reached 72.2%, indicating that approximately 82.01 million individuals have internet access. This data was published in December 2024 by the Ministry of Communications and Information Technology (MCIT) in their “ICT Indicators in Brief” report. For reference, the report can be accessed at the following link: https://mcit.gov.eg/en/Publication/Publication_Summary/10552

3.3 Are there national research and education networks (NRENs) active in the country?

**YES**, Egypt has a robust National Research and Education Network (NREN) that connects various educational and research institutions across the country. This network serves as a vital open infrastructure for promoting collaboration, resource sharing, and advanced research activities. This comprehensive national network ensures that students, educators, and researchers across different levels of education and research can access and share resources, collaborate on projects, access Egyptian Knowledge Bank educational & research resources and engage in advanced research activities. The network facilitates seamless communication and data exchange, fostering a vibrant educational and research ecosystem. Through the Egyptian National STI Network, Egypt's NREN is connected to global Research and Education Networks, including (Internet2, and GÉANT Network) and is planned to be connected through MEDUSA 100Gbps Cable, funded by European Investment Bank. These connections are crucial for enabling Egyptian researchers and educators to participate in international collaborations, access global research resources, and contribute to the global knowledge community. By linking with networks like Internet2 and GÉANT, Egypt ensures high-bandwidth connectivity and access to advanced network services, which are essential for modern research and education.

3.4 Are there national or institutional open science infrastructures in your country?

**Yes**, Egypt has established several significant national and institutional open science infrastructures. These infrastructures support various aspects of open science, from knowledge access and data sharing to collaborative research environments. Key Open Science Infrastructures in Egypt:

**Egyptian Knowledge Bank (EKB):**

- The EKB serves as a national digital library and knowledge hub, providing open access to a wealth of resources, including scientific research, educational materials, and cultural content.
- It acts as a centralized platform for accessing and sharing scientific knowledge, promoting discoverability and responsible reuse.
- The EKB also plays a crucial role in capacity building and fosters collaboration among researchers.

**Egyptian Open Access Publishing Platform:**

Hosted locally at STGC[3] Datacenter, with over 1040 journals and more than 390,000 articles, this platform provides a robust infrastructure for disseminating Egyptian research globally. Its success is evident in over 200 million global downloads in just five years, with users from the USA, Saudi Arabia, Algeria, and Iraq following Egypt in accessing the platform. Crucially, it supports multilingual scientific knowledge, including 483 Arabic journals. Furthermore, the Egyptian Open Access Publishing platform have received over 3.2 million citations from international Scopus-indexed journals, this demonstrates the quality of the published articles across the platform. Egypt's Open Access Publishing Platform welcomes submissions from non-Egyptian authors, with over 1.8 million authors contributing, emphasizing Egypt's belief in borderless scientific collaboration.

- This platform provides a robust infrastructure for disseminating Egyptian research globally.
- It hosts a large number of journals and articles, supporting multilingual scientific knowledge.
- The platform's success is evident in its global downloads and citations, demonstrating the reach and impact of Egyptian research.
- **Transformative Agreements:** Egypt's transformative agreements, such as the landmark agreement with Springer Nature, provide Egyptian researchers with opportunities to publish in leading international journals while ensuring open access. From 2022-2024, Egyptian researchers published 9,315 articles through the Springer Nature agreement, receiving 47,497 citations, 11,161 Altmetrics, and 17,288,599 global downloads. Global readership is diverse, spanning Asia (55%), North America (14%), Europe (13%), Africa (13%), South America (2%), and Australia (1%).
- **Supporting Local Journals Transformation to OA Model:** The Academy of Scientific Research and Technology (ASRT) has for over a decade supported local Egyptian journals in transitioning to open access models through partnerships with leading publishing houses like Elsevier, Taylor & Francis, Wolters Kluwer, and Digital Commons. This initiative has facilitated the publication of 132,244 articles across 86 supported journals, and enabling their inclusion in international abstract and citation indexes. This capacity building has directly contributed to the success of the creating and operating the Egyptian open access publishing platform.

**Egyptian Knowledge Graph (EKG):**

- Developed as a comprehensive big-data research performance analytics platform, the EKG is a key open science infrastructure.
- It evaluates the entire Egyptian research output, including for the first time the non-indexed locally published as well as indexed articles, regardless of language.
- The EKG facilitates open evaluation, integration, and analysis, of the wider corpus of Egyptian research output, supporting evidence-based policymaking and research assessment.

**National Science & Technology Grid & Cloud Computing Centre (STGC[3]):**

- Hosted at ASRT, this centre houses critical open science infrastructures, including the EKB, the national publishing platform, and the EKG.
- It provides high-performance computing capabilities, supporting advanced research in areas like precision medicine and genetics.
- When discussing the "EKG" in the context of Egyptian research output, it's crucial to emphasize its potential to provide a truly comprehensive view, especially in areas often overlooked by traditional citation analyses.
- Global citation analyses often heavily favour publications in English within STEM fields. This can lead to an underrepresentation of research in the social sciences and humanities, where local languages and contexts play a significant role.

- The EKG aims to address this by encompassing a broader range of research outputs, including those in Arabic, thus providing a more accurate and equitable reflection of Egypt's scholarly contributions.

**Open Educational Resources (OER) Platforms:**

- Egypt has launched open education platforms for Egyptian K-12 students (Study platform) with a wide range of learning objects aligned with the new curriculum.
- A national LMS for secondary schools and higher education, with the source code openly available for universities to customize.

**Open Infrastructure through ENSTINET:**

- Egypt has established robust open infrastructure, connecting to major Global Research & Education Networks (Internet2 and GEANT) through the Egyptian National STI Network (ENSTINET), extending access to all public universities, research centres, and 3000 secondary schools.

**National Bank for Scientific Laboratories and Equipment (NBSLE):**

Initiated by the SCU in 2015, the NBSLE aims to optimize the utilization of scientific resources within Egyptian universities. Its primary objectives are:

- **Comprehensive Database Creation**: Developing an up-to-date information system cataloging scientific laboratories and equipment across universities. This database facilitates researchers' access to essential tools and promotes collaborative use of resources.
- **Enhanced Research Efficiency**: By providing detailed information on available equipment and facilities, the NBSLE streamlines research planning and execution, reducing redundancy and fostering innovation he collaborative efforts of the SCU and NBSLE underscore Egypt's commitment to advancing its scientific and academic sectors. By ensuring standardized education and optimizing research resources, these institutions contribute significantly to the nation's educational and technological progress.

These infrastructures collectively contribute to creating a more open, accessible, and collaborative scientific environment in Egypt, aligning with the principles and goals of Open Science.

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science?

**Yes,** Egypt actively funds and collaborates on federated regional and international information technology infrastructure for Open Science. A notable example of this is Egypt's funding and partnership with Clarivate Analytics to launch and support the Arabic Citation Index (ARCI).

**Arabic Citation Index (ARCI):**

Egypt has played a crucial role in the establishment and funding of the Arabic Citation Index (ARCI). This initiative is a significant step towards enhancing the visibility and accessibility of Arabic scientific research on a global scale. Key aspects of the project include:

- **Funding and Support:** Egypt provides financial support and resources for the development and maintenance of the ARCI. This funding ensures the index's sustainability and continuous improvement.
- **Regional Impact:** The ARCI is a regional initiative that indexes and evaluates Arabic scientific output from 22 Arab states. This effort helps to bridge the gap in access to Arabic research, making it more discoverable and accessible to researchers worldwide.
- **Enhanced Visibility:** By indexing Arabic scholarly content, the ARCI increases its visibility and impact on the global stage. This helps to ensure that Arabic research is recognized and considered in international academic discourse.
- **Language Barrier:** The ARCI addresses the language barrier by indexing Arabic publications and providing translated abstracts. This makes it easier for non-Arabic speaking researchers to identify and understand important Arabic research.

**Open Access Initiatives:**

- Through its open access publishing platforms, the national project to develop and transition Egyptian journals to open access models with leading publishers, and the open access transformative agreement with Springer Nature, Egypt contributes to the global open access landscape.

- With over 1040 journals and more than 390,000 articles, this platform provides a robust infrastructure for disseminating Egyptian research globally. Its success is evident in over 200 million global downloads in just five years, with users from the USA, Saudi Arabia, Algeria, and Iraq following Egypt in accessing the platform. Crucially, it supports multilingual scientific knowledge, including 483 Arabic journals. Furthermore, the Egyptian Open Access Publishing platform have received over 3.2 million citations from international Scopus-indexed journals, this demonstrates the quality of the published articles across the platform.

- **Egyptian Knowledge Bank Int.:**
  - Egyptian Knowledge Bank International (EKB Int.) initiative, where Egypt is planning to expand the services of EKB regionally through partnering with Association of Arab Universities and Federation of Arab Scientific Research Councils, in collaboration with 15 international publishers and technology providers.
- **Egyptian National STI Network (ENSTINET):**
  - ENSTINET serves as the backbone for connecting Egyptian research and education institutions to global networks. It acts as a crucial link to international infrastructure.
  - **Connectivity:** ENSTINET connects Egyptian universities, research centers, and schools to global Research & Education Networks such as Internet2 and GÉANT. This connectivity allows for the exchange of data, participation in collaborative projects, and access to international resources.
  - ENSTINET-NREN is a member of the Arab States Research & Education Network (ASREN), sharing and collaborating on the establishment of a regional Open Science Platform.
  - **Scope:** ENSTINET facilitates access to a wide range of research areas, supporting all elements of Open Science, including data sharing, open access publications, and collaborative tools.
  - **Accessibility:** ENSTINET aims to provide broad accessibility to researchers, educators, and students across Egypt, ensuring they can engage in international scientific activities.

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms?

**Yes,** Egypt's involvement in North-South, North-South-South, and South-South collaborations to optimize infrastructure use can be exemplified by its partnerships with various countries and organizations.

- **North-South Collaboration:** Egypt's collaboration with the European Union on research and innovation projects, including those related to open science, connectivity, exemplifies North-South cooperation.
- **North-South-South Collaboration:** Egypt's participation in initiatives led by international organizations like the World Bank, where it collaborates with other developing countries and developed nations, can be seen as North-South-South collaboration.
- **South-South Collaboration:** Egypt’s partnerships with other African countries to share knowledge and resources related to open science and research infrastructure demonstrate South-South collaboration.

- **Some Examples on International Collaboration:**
  - Egypt engages in North-South, North-South-South, and South-South collaborations to optimize infrastructure use. This includes joint strategies for shared, multinational, regional, and national Open Science platforms.
  - Egypt has joined the EUMEDPlus connectivity project to connect Euro Mediterranean Countries and share research & open infrastructure resources and to encourage cross borders collaboration.
  - Participating in the planned MEDUSA connectivity project that will connect research and education entities across the mediterranean countries with 100Gbps.

- Alignment and particpation with Arab States Research & Education Network towards open science platform.
- **Egyptian Knowledge Bank Int.:** Egyptian Knowledge Bank International (EKB Int.) initiative, where Egypt is planning to expand the services of EKB regionally through partnering with Association of Arab Universities and Federation of Arab Scientific Research Councils, in collaboration with 15 international publishers and technology providers.
- **Egyptian Open Access Publishing Platfom:**
  Through its open access publishing platforms, which encompass over 1040 journals and more than 390,000 articles, this platform provides a robust infrastructure for disseminating research globally. It offers opportunities for both north-south and south-south scientific collaboration in the creation of science through co-authoring and publishing openly on the platform. Its success is evident in over 200 million global downloads in just five years, with users from the United States, Saudi Arabia, Algeria, and Iraq following Egypt in accessing the platform. Crucially, it supports multilingual scientific knowledge, including 483 Arabic journals. Furthermore, the Egyptian Open Access Publishing platform has garnered over 3.2 million citations from international Scopus-indexed journals, which underscores the quality of the published articles across the platform.
- **International Practices & Collaboration:** Egypt, as a member of both the AU and an active participant in BRICS-related initiatives, aligns with these policies, integrating open science principles into its national research frameworks to foster greater collaboration and knowledge sharing across Africa and the broader international community. Both the BRICS nations and the African Union (AU) have been actively promoting open science policies to enhance collaborative research and innovation. The BRICS countries with Egypt one of the state members, have developed unique approaches to open access, blending public investments and private initiatives to foster research as a driver of economic and societal development. Collectively, they emphasize open science to improve the quality, impact, and accessibility of scientific information. In a significant move, China, Brazil, South Africa, and the AU jointly launched the "Initiative on International Cooperation in Open Science," aiming to create an equitable and inclusive global environment for scientific and technological advancement, particularly benefiting nations in the Global South. The AU's commitment is further reflected in its Science, Technology, and Innovation Strategy for Africa 2024 (STISA-2024), which underscores the importance of open science in addressing the continent's socio-economic challenges.

These collaborations often involve sharing open science infrastructures, providing technical assistance, and exchanging best practices.

# 4. Investing in human resources, training, education, digital literacy and capacity building for open science

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country?

**Yes,** Open Science capacity building is being systematically developed in Egypt's higher education and research institutions through various initiatives and programs that promote and embed Open Science principles and practices.

**Egypt's Capacity Building in Open Science Practices and Principles**

- **Learning by Practice Through Open Access Initiatives:** Egypt's commitment to open access publishing increases the visibility and impact of Egyptian research, contributing to capacity building by making research outputs more widely available and promoting knowledge exchange. Institutions are building capacity in open science publishing workflows and practices by encouraging researchers to publish in open access formats.
- **Training and Workshops:** Training and workshops on utilizing the Egyptian Knowledge Bank (EKB) and the Egyptian Knowledge Gateway (EKG) platforms and adhering to open data principles are actively taking place, enhancing the skills and knowledge of researchers and students in open science practices.

- **Professional development programs and teaching excellence programs delivered through EKB network of partners**, incorporate open science principles in education and research.
- The ongoing ASRT training workshops as part of the national initiative to develop, publish, and transition Egyptian journals to Open Access models with leading publishers.
- Promoting open access through training programs for editors-in-chiefs, and authors.
- Training on scientific manuscripts writing, peer-reviewing, plagiarism, open metrics, and related topics, encourages open science and scientific integrity awareness across the community.
- **Collaboration with International Partners:**Egypt's collaboration with international organizations and networks in science and technology facilitates capacity building through joint projects, training programs, and knowledge exchange. These collaborations expose Egyptian researchers to international best practices in Open Science, thereby enhancing their skills and knowledge.

4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

While explicit, standalone capacity-building programs specifically targeted at policymakers on integrating Open Science core values and principles may not be extensively developed, there are underlying efforts and structures in place that contribute to this goal. Egypt's approach to promoting Open Science and its values involves multiple facets:

**Integration within STI Initiatives:**

Egypt's Science, Technology, and Innovation (STI) policies and strategies, such as the National STI Strategy, indirectly promote Open Science principles & values. While these policies may not always explicitly use the term "Open Science," they emphasize principles such as transparency, collaboration, and knowledge sharing. Policymakers involved in the development and implementation of these policies are thus engaged with these concepts.

**Role of ASRT and STDF:**

Organizations like the Academy of Scientific Research and Technology (ASRT) and the Science & Technology Development Fund (STDF) play a crucial role in promoting research and innovation. Their activities often involve workshops, seminars, and conferences that bring together researchers, policymakers, and other stakeholders. These events serve as informal capacity-building opportunities for policymakers, exposing them to Open Science principles and best practices.

**International Collaborations:**

Egypt's participation in international collaborations in science and technology provides avenues for policymakers to learn about Open Science policies and practices in other countries. These collaborations often involve exchanges of expertise and best practices, contributing to the knowledge and awareness of policymakers.

Egypt recognizes the crucial need to invest in capacity-building programs specifically tailored for policymakers. By equipping decision-makers with a deep understanding of Open Science's core values and principles, we empower them to create a conducive policy environment that fosters collaboration, transparency, and accessibility in research.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025?

While efforts have been made to modernize higher education curricula in Egypt, particularly in integrating technological knowledge and aligning with sustainable development goals, fully incorporating open science competencies framework in research skills curricula **within the Egyptian higher education landscape still faces challenges, primarily due to Resource Limitations and sustainable funding**. These limitations encompass

infrastructure, training, and financial investment, hindering the development and sustainability of open science programs.
Actions Taken by the Egyptian Ministry of Higher Education & Scientific Research to incorporate the framework of open science competencies in research skills curricula.

- Actively enhance awareness and understanding of Open Science to facilitate its timely adoption and maximize its impact.
- Embed Open Science principles across all stages of the scientific process, ensuring accessibility, collaboration, and knowledge sharing.

4.4 Are open educational resources as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above?
PARTLY

4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large?

**Yes,** Egypt has several initiatives aimed at supporting science communication in line with Open Science values, facilitating the dissemination of scientific knowledge to various audiences, including researchers, policymakers, and the public. Here are some key examples:
**Egyptian Knowledge Bank (EKB):**

- The EKB is a significant national initiative that provides access to a vast collection of scientific information and resources. By making these resources openly accessible, the EKB facilitates the dissemination of scientific knowledge to a wide audience.
- It serves as a central platform for researchers, educators, students, and the public to access and engage with scientific content.
- The EKB plays a crucial role in bridging the gap between scientific knowledge and its application in various sectors of society.
- **Notable global dissemination of quality scientific knowledge:** Egypt demonstrates strong commitment to open access publishing, an analysis of Scopus-indexed research manuscripts from Egypt between 2019-2024 reveals a significant contribution to open access (OA) publishing. Of the 230,970 indexed manuscripts, 117,871 (51%) were published as OA, with 83,861 (36%) specifically being Gold OA.
- However, this data only represents a portion of Egypt's OA output. A broader analysis, incorporating non-Scopus indexed OA articles, shows an even stronger commitment to OA. For instance, the Egyptian Open Access Publishing Platform alone facilitated the publication of over 210,000 Gold OA peer-reviewed articles during the same period.
- Considering this wider scope, and by eliminating the number of articles that are in both Scopus and EKB of 12,490 articles during the same period, the overall percentage of OA articles published by Egyptian researchers between 2019-2024 reaches an impressive 74%, with Gold OA accounting for 67% of this total. This demonstrates Egypt's leading role in promoting and embracing open access and knowledge dissemination to researchers), the open access nature of the national publishing platform and transformative agreements, and the global readership demonstrate a commitment to open science values. Egypt’s Open Access Publishing Platform welcomes submissions from non-Egyptian authors, with over 1.8 million authors contributing, emphasizing Egypt's belief in borderless scientific collaboration.
- **Open Access Performance Metrics:** Egypt's OA publications demonstrate high quality and impact (Source: Web of Science):
  - Category Normalized Citation Impact: 1.339 (34% higher than global average)
  - Q1 and Q2 Indexing: 79% (higher than global average of 75.6%)
  - Percentage of Cited Documents: 77.49% (surpassing all countries relative to publication volume).

**Science Communication Activities by ASRT and STDF:**

- The Academy of Scientific Research and Technology (ASRT) and the Science & Technology Development Fund (STDF) frequently organize workshops, conferences, and public events aimed at disseminating scientific knowledge and promoting public engagement with science.
- These events often involve presentations, discussions, and demonstrations that make scientific concepts more understandable and relatable to the general public.

**Open Science Community Egypt (OSCE):**

- Established in 2023, Open Science Community Egypt (OSCE) is the country's first network dedicated to promoting open science frameworks. Founded by experienced Egyptian professionals, OSCE focuses on fostering a culture of openness and collaboration within the research community.

**Open Educational Resources (OER):**

- Egypt has launched open education platforms for Egyptian K-12 students (Study platform) with a hunderes of thousands of learning objects aligned with the new national curriculum. A national LMS for secondary schools and universities. This makes educational content widely accessible and promotes the dissemination of knowledge.

**Media and Public Outreach:**

- Egyptian media outlets often feature reports and stories about scientific research and innovation. This helps to raise public awareness of scientific developments and their potential impact on society.
- Researchers and scientists are increasingly encouraged to engage with the media and participate in public outreach activities to communicate their findings to a broader audience.

Egypt has several initiatives that support science communication in line with Open Science values and principles. The Egyptian Knowledge Bank and Egyptian Innovation Bak, play central roles in making scientific knowledge and data accessible, while organizations like ASRT and STDF facilitate public engagement through various events and activities. Additionally, the use of Open Educational Resources and media outreach efforts contribute to the broader dissemination of scientific knowledge. These initiatives collectively help to ensure that scientific information reaches researchers, decision-makers, and the public, promoting a more informed and engaged society.

## 5. Fostering a culture of open science and aligning incentives for open science

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025?

Egypt has successfully initiated the groundwork and solid foundation for open science. To fully realize its potential, the integration of open science principles within research assessment and evaluation processes should be prioritized and remains an area for potential development.

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025?

Despite specific clauses recognizing Open Science practices in career evaluation and progression systems are not yet widely implemented in Egypt, traditional metrics, such as publications in high-impact journals, often dominate evaluation processes. However, there is a clear direction towards adopting these principles. With continued efforts and alignment with the UNESCO Open Science Recommendation, it is anticipated that career evaluation systems will increasingly reflect the values of openness, collaboration, and engagement in research. The Egyptian Knowledge Bank, Egypt Open Access Publishing Platform, the Egyptian Knowledge Graph (Big-Data Research Analytics & Inights Platform), the Egyptian Innovation Bank, and the National STI Strategy provide strong foundations for this transition. The direction and momentum indicate a strong likelihood of increased recognition of open science practices within career evaluations in the near future. The following programs and initiatives could serve as a suitable starting point for enhancing recognition of open science practices within career progression and evaluation:

- The Egyptian Ministry of Higher Education and Scientific Research has introduced several initiatives to enhance career development and evaluation within the higher education sector: **"Be Ready" Initiative**: Launched in collaboration with the International Labour Organization (ILO) and the UK's Department for International Development (UK AID), this program aims to equip youth and recent graduates with skills aligned to the labor market. The initiative focuses on vocational guidance, developing job service platforms, and enhancing youth capabilities to secure better employment opportunities. It seeks to train over 2,000 young individuals and create more than 1,000 decent jobs by the end of the year.
- Initiative from **Supreme Council regarding Career path Conversion** is proposed by the end of 2024. This includes Non-Degree Credentials (NDCs) include a wide array of credentials beyond academic degrees conferred by accredited educational and training institutions. NDCs are generally both more affordable and shorter in duration than traditional degree pathways. They are also attractive options for individuals looking to quickly upskill or reskill in a particular field. NDCs will play a vital role in helping individuals become more employable and subsequently fill existing gaps in the labor market. NDC includes Apprenticeship (Degree apprenticeships are developed between the Government, employers and universities, for learning skills that will be in high demand in the future), Vocational training (Vocational Courses are educational programs that are designed for specific careers or trades. They typically focus on practical skills and knowledge rather than theoretical or academic concepts). Micro-creditional (Certificates are issued upon the completion of a course or series of micro-credential courses covering a specific skill set or body of knowledge). Bootcamps (A bootcamp is a specialized, intensive training program designed to rapidly increase or significantly affect a participant's knowledge, abilities, or both). Certificate programs (Certifications awarded by professional organizations to signify that an individual has reached a specific level of skill or competence in a particular field). Industry certification, Professional licenses. These programs focus on elevating technical skills. Also, conversion programs are proposed via the initiative. Conversion courses are intensive postgraduate programs that allow pursuing a career in a major that does not match undergraduate degree, or professional career has not prepared.
- The newly announced reference framework for higher education by SCU introduces a small thesis course scheme. Through this approach some of the offered courses can support open science by creating a research team of academic staff, undergraduate students and postgraduate students. This can support the sharing of knowledge about ongoing research projects and related publications thus creating a supportive environment for open science. The reference framework also offers a ‘minors’ track which accepts registration of those who are interested in developing skills related to the job market in fields other than their registered study scopes. This enhances a multidisciplinary approach to education and learning and eases students' entry to the job market. The reference framework introduces a ‘selected topics’ course scheme which is meant to be flexible to match recent research topics and advance the students’ analytical and research skills. It can be introduced in consecutive levels to support both the students research skills and skills related to job market entry based on most recent academic research in the field of study.
- In general, Egyptian research is increasingly driven by societal needs, aligning with national challenges as reflected in the SDG research mapping. This mapping highlights that 16 of the SDGs were addressed in scientific publications, with several surpassing the global average in terms of field-weighted citation impact. Some even exceeded it by more than twice the global average. This demonstrates strong engagement and impact not only in quantity but also in quality, as evidenced by exceeding the global average citation impact. The top Sustainable Development Goal (SDG) in terms of research outputs is also aligned with key national priorities, such as SDG 3 (Good Health and Well-being), SDG 7 (Affordable and Clean Energy), and SDG 6 (Clean Water and Sanitation).

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025?

Egypt has established the groundwork for recognizing and rewarding open science practices through its investment in open science supporting national knowledge infrastructure. Although explicit reward mechanisms may currently be limited, there are strong indications that such mechanisms will evolve. The existing platforms and strategies provide a foundation for future policies and initiatives that will more directly acknowledge and incentivize open science. It is reasonable to anticipate more concrete steps towards integrating open science into research evaluation and reward systems in Egypt.

In this regard, there are several implicit recognitions currently in place, such as:

**Egyptian Knowledge Bank (EKB) & Egypt Open Access Publishing Platform:**

The establishment of the EKB can be seen as an implicit form of recognizing and rewarding open science. By creating national platforms for open access to knowledge and data sharing, Egypt is:

- Facilitating wider dissemination of research, which can enhance researchers' visibility and impact.
- Supporting collaborative research, which can lead to greater recognition and rewards in the scientific community.
- Improving the overall research ecosystem, which indirectly benefits researchers who embrace open science practices.

**Egyptian Knowledge Graph (EKG):** a more inclusive Big-Data Research Analytics & Insights Platform implicitly recognizes and rewards open science practices in Egypt in several ways:

- **Comprehensive Data Aggregation:**The EKG aggregates research information from various sources, including local and international publications, patents, funding data, and preprints. This comprehensive approach supports the open science goal of sharing diverse research outputs.
- **EKG recognition for marginalized research communities:** EKG leveraged to democratize expert recognition and discovery, particularly for marginalized research communities in social sciences and humanities. This platform transcends traditional, English-centric evaluation by incorporating research published in Arabic and local journals, effectively dismantling language barriers. By leveraging EKG-driven inclusive analysis, Egypt will provide a more just and inclusive system for identifying and celebrating the contributions of these vital experts, ensuring their knowledge is valued and integrated into the broader research community.
- **Emphasis on Collaboration:** The platform encourages the formation of research workgroups and teams, fostering national, regional, and international collaborations. This aligns with open science's emphasis on collaborative research.
- **Open Access Support:** The platform's support for upgrading local journals and assisting with applications for international indexing, indirectly promote open access publishing.
- **Data-Driven Decision Making:** By providing accurate and reliable data, the EKG enables evidence-based decision-making, which can lead to policies and initiatives that further support open science practices.
- **Increased Visibility:** The platform aims to identify and recognize researchers with active collaborative research works on both local and global rearch activities . This aligns with open science principles, which advocate for furthering research collaboration to a wider audience.

**Arabic Citation Index (ARCI):**The also serves as an implicit recognition mechanism by:

- Increasing the visibility of Arabic scholarly content, which can enhance researchers' standing and influence.
- Promoting scholarly communication and knowledge sharing within the Arabic-speaking world.

**STI Strategy:** The Egypt National STI Strategy include provisions that indirectly reward open science practices through:

- Funding priorities for collaborative research.
- Recognition of research that addresses societal needs and promotes knowledge sharing.

- Support for research infrastructure that enables open access.

**Capacity Building and Training Programs:** As open science practices become more established, there are increased training and capacity building programs through EKB, which serve as a form of recognition by:

- Providing certifications or credentials for open science skills.
- Highlighting researchers who participate in and contribute to open science training initiatives.

5.4 Have there been any unintended negative consequences of open science practices reported in your country?

**Yes, The Challenge of Costs and Sustainability in Open Access Publishing:**
Open access publishing can be expensive due to article processing charges (APCs), which can create barriers for researchers in developing countries. Egypt has taken steps to alleviate this burden through three key initiatives:

- A national project to support Egyptian journals' transition to open access, funded by the Academy of Scientific Research & Technology (ASRT).
- A transformative agreement between the Science and Technology Development Fund (STDF), the Egyptian Knowledge Bank (EKB), and Springer Nature.
- The Egyptian Open Access Publishing platform, hosting over 1,000 journals, operated by ASRT in collaboration with EKB.

However, maintaining government funding for these initiatives amidst economic challenges can be difficult. This can lead to unequal access to publishing and hinder research dissemination, particularly from developing regions. While existing agreements and initiatives aim to address this issue, further action is needed to ensure equitable & sustainable open access publishing opportunities for researchers in Egypt and beyond.

# 6. Promoting innovative approaches for open science at different stages of the scientific process

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process?

Egypt has several initiatives at both the national and institutional levels that promote innovative, participatory changes in the research cycle to enhance openness. These initiatives address various recommended actions, including:

1. **Promoting Open Innovation Practices:**
   - **Egyptian Innovation Bank (EIB):** The EIB encourages open innovation practices by creating a platform for engagement between the public, researchers, academics, and industry partners. This fosters collaboration and the development of new solutions and products.
   - By involving diverse stakeholders, the EIB promotes an open and inclusive innovation ecosystem, which is crucial for driving scientific progress.
2. **Participatory Methods and Citizen Science:**
   - The EIB's platform implicitly supports participatory methods by involving societal actors in the innovation process. The EIB's approach aligns with the principle of engaging the public in knowledge generation and problem-solving (Citizen Science).
3. **Egyptian Publishing Platform and Open Peer-Review:**
   - The Egyptian Publishing Platform facilitates the dissemination of research and encourages open access. The platform's accessibility inherently promotes broader review and engagement with international research communities.
   - The platform's open access nature allows for wider scrutiny and feedback on research, which can be seen as an element of open peer review.
4. **Development of Egyptian Knowledge Graph (EKG):**
   - The EKG promotes a more open research valuation and analytics system. By providing comprehensive data and insights on research performance, the EKG supports the recognition and valuation of diverse research outputs, including those from marginalized communities in social sciences and humanities, as well as research in Arabic language.
   - The EKG helps address biases in research evaluation and promotes a more inclusive approach to assessing scientific impact.

5. **Arabic Citation Index (ARCI):**

- The ARCI supports inclusive visibility of Arabic scientific literature from all Arab League countries. By indexing and evaluating Arabic research, the ARCI ensures that this body of knowledge is discoverable and recognized in the global scientific community.
- This initiative directly addresses the need for inclusivity in scientific communication, ensuring that research in Arabic is not marginalized or overlooked.

**Conclusion:**

Egypt has implemented several initiatives at national and institutional levels to promote open science practices and enhance the transparency and inclusivity of the research cycle. The Egyptian Innovation Bank, Egyptian Publishing Platform, Egyptian Knowledge Graph, and the Arabic Citation Index all contribute to creating a more open and participatory scientific environment. These efforts align with the principles of open innovation, citizen science, open peer-review, and inclusive research evaluation, as outlined in the UNESCO Open Science Recommendation.

# 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science?

**Yes,** Egypt actively promotes and facilitates international scientific collaborations on open science through several key initiatives, demonstrating a strong commitment to the principles outlined in the 2021 UNESCO Recommendation on Open Science.

**Egyptian Knowledge Bank International (EKB Int.) Initiative:**

- The EKB is expanding its services regionally and internationally, following UNESCO commendation.
- Egypt is partnering with the Association of Arab Universities and the Federation of Arab Scientific Research Councils to collaborate on expanding EKB services and the Open Access Publishing platform regionally.
- This initiative aims to share resources, enhance access to knowledge, and foster collaboration among researchers across the Arab region.

**Regional Knowledge Graph Development:**

- Egypt is planning to develop a regional knowledge graph based on the groundbreaking Egyptian Knowledge Graph. This comprehensive and extensive knowledge graph will serve as a Big Data Platform specifically designed for research analytics and insights.
- This platform is intended to evolve into a regional Knowledge Graph, enabling broader data sharing and analysis across borders.
- By developing this regional Knowledge Graph, Egypt aims to facilitate international collaboration on data-driven research and insights.

**Arabic Citation Index (ARCI):**

- Egypt funds and collaborates with Clarivate Analytics on the Arabic Citation Index (ARCI).
- ARCI indexes and evaluates Arabic scientific output from 22 Arab states, enhancing the visibility and accessibility of Arabic research globally.
- This initiative promotes international collaboration by making Arabic research discoverable and integrated into the global scholarly community.

**Global Research & Education Networks:**

- Egypt connects to global Research & Education Networks such as GÉANT and Internet2 through the Egyptian National STI Network (ENSTINET).
- These connections facilitate Egyptian researchers' participation in international collaborations, granting them access to global research resources. They also enable participation in research projects that require high-bandwidth connectivity, such as

Egyptian research organizations' involvement in CERN research activities. These contributions collectively contribute to the global knowledge community.

- Participation in projects like MEDUSA further strengthens these connections and facilitates high-bandwidth data exchange for collaborative research.

**International Agreements and Partnerships:**

- Egypt engages in international agreements for joint research and open science with various countries and organizations.
- These partnerships often involve collaborative projects, data sharing initiatives, and capacity-building programs.
- Such collaborations enhance research capabilities, foster knowledge exchange, and promote open science practices on a global scale.

These initiatives demonstrate Egypt's commitment to promoting and facilitating international scientific collaborations on open science. By investing in infrastructure, platforms, and partnerships, Egypt is creating an environment that supports open and collaborative research on a global scale.

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science?

**Yes,** Egypt has a clear strategy and active plans for stimulating cross-border multi-stakeholder collaboration on Open Science. This involves several key initiatives and partnerships:

**UNESCO-UNICEF Gateways to Public Digital Learning Initiative:**

- Egypt actively participates in the UNESCO-UNICEF Gateways to Public Digital Learning Initiative.
- In May 2024, Egypt facilitated a case study visit with the participation of 21 countries. This visit focused on sharing best practices and lessons learned in developing, enriching, and sustaining the Egyptian Knowledge Bank (EKB).
- This initiative demonstrates Egypt's commitment to collaboration and knowledge exchange in the realm of digital learning and open access.

**Egyptian Knowledge Bank International (EKB Int.) Initiative:**

- The EKB is expanding its services regionally and internationally, building upon UNESCO's commendation.
- Egypt is partnering with the Association of Arab Universities and the Federation of Arab Scientific Research Councils to extend EKB services and the Open Access Publishing platform across the Arab region.
- This initiative aims to foster collaboration, enhance access to knowledge, and share resources among researchers in the Arab region.

**Regional and Global Research & Education Networks:**

- Egypt is connected to regional and global Research & Education Networks, including GÉANT and Internet2, through the Egyptian National STI Network (ENSTINET).
- These connections facilitate international collaborations, data exchange, and access to global research resources.
- Participation in projects like MEDUSA further enhances these connections, enabling high-bandwidth data transfer for collaborative research.

**Arabic Citation Index (ARCI):**

- Egypt collaborates with Clarivate Analytics on the Arabic Citation Index (ARCI), which indexes and evaluates Arabic scientific output from 22 Arab states.
- This initiative promotes international collaboration by making Arabic research discoverable and accessible to the global scientific community.

**Capacity Building and Best Practices Exchange:**

- Egypt engages in various capacity-building programs and initiatives in collaboration with regional partners such as Association of Arab Universities, and Federation of Arab Scientific Research Councils, to deliver training programs in collaboration with EKB network of international partners.
- These programs focus on training researchers, educators, and policymakers on Open Science/Open Education.

- Egypt actively exchanges best practices with other countries through conferences, workshops, webinars, and joint projects.

Through these multifaceted strategies and initiatives, Egypt actively promotes and facilitates cross-border multi-stakeholder collaboration on Open Science. This commitment is evident in its participation in global initiatives, regional partnerships, and the development of open infrastructure and platforms. Egypt aims to contribute to a more open, collaborative, and inclusive global scientific landscape.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science?

**Yes,** Egypt is actively involved in discussions on the establishment of regional and international funding mechanisms for Open Science implementation. This engagement is primarily driven by the ambition to expand access to knowledge, promote research collaboration, and ensure the sustainability of open science initiatives.

**Egyptian Knowledge Bank International (EKB Int.) and Partnerships:**

- **EKB Expansion:** The Egyptian Knowledge Bank International (EKB Int.) initiative is a key driver for Egypt's involvement in these discussions. The EKB is expanding its services regionally and internationally, and this expansion necessitates exploring sustainable funding mechanisms.
- **Partnership with AArU and FASRC:**Egypt has established partnerships with the Association of Arab Universities (AArU) and the Federation of Arab Scientific Research Councils (FASRC). These partnerships aim to collaborate on expanding EKB services and the Open Access Publishing platform regionally.
- **ASREN Partnership:** Egypt is also partnering with the Arab States Research and Education Network (ASREN) to explore opportunities and funding mechanisms for an open science platform.
- **Objectives:** These collaborations seek to identify and develop funding models that can support the establishment and maintenance of regional open science platforms, ensuring their long-term sustainability and accessibility.

## 8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

**Yes, The Challenges of Costs and Sustainability in Open Science (Access, and Infrastructure) and the Less Equitable and Less Inclusive Evaluation of Scientific Contributions, in other words, the Lack of a World-Recognized Open Metrics Platform, as Expressed in the Following Integral Part of Egypt's Report, Are Pertinent Concerns That Impede the Sustainable Implementation and Realization of Values for All:**

The UNESCO Recommendation on Open Science is undoubtedly a landmark achievement, illuminating a pathway towards a more equitable and accessible scientific landscape. However, its true potential hinges on addressing the persistent challenges that threaten to perpetuate existing disparities, particularly for the Global South.

The transition from subscription-based models to open access, while well-intentioned, has inadvertently created a new set of hurdles. The steep Article Processing Charges (APCs) imposed by many publishers, primarily based in the Global North, are proving to be a significant barrier to participation for researchers in developing countries (mainly Global South); major publishers are proxying Open Science/Access for adding more revenues. This mirrors the historical imbalance and inequity in which the Global North reaped disproportionate benefits from subscription fees through publishing houses located primarily in the Global North, while the Global South struggled to access essential knowledge. The current reality, as evidenced by the usage patterns of Egyptian open access platforms and transformative agreements, reveals a concerning trend: despite the financial burden borne by Egypt, "a Global South nation", the primary beneficiaries of open access outputs remain institutions and researchers in the Global North in term of access and downloads. This disparity extends beyond access to the creation of science, as evidenced by the high cost of APCs, effectively restricting participation from Global South researchers.

Furthermore, the lack of universally accepted and equitable more inclusive open metrics, that is currently dominated by entities from the Global North, particular publishing houses, raises concerns about possible bias and conflicts of interest. This control over science valuation further disadvantages researchers & organizations in developing countries, and with world institution rankings are primarily based on these metrics that are less inclusive even in terms of language representation “which index and favor English-Language research outputs”.

The emergence of generative AI adds another layer of complexity. The Global North, with its superior resources and infrastructure, is poised to capitalize on open science outputs, including those from the Global South, to train advanced AI models openly. These models, often offered through expensive subscription services, will likely exacerbate the existing knowledge and technology gap, creating an "AI divide" that further marginalizes developing nations.

Therefore, we request & urge UNESCO to move beyond simply monitoring the implementation of Open Science Recommendation and to actively engage in supporting its realization. This requires concrete actions, including:

- **Establishing a global, open, and equitable metrics platform, democratizing science evaluation:** potentially integrated with and based on initiatives such as OpenAlex and/or DOAJ, to ensure fair, and more inclusive holistic evaluation of scientific contributions. OpenAlex provides additional coverage for the humanities, non-English languages, and the Global South, with over 250 million scholarly works, (more than twice the number of indexed works over reputable citation indexes), 90 million disambiguated authors, (more than four times the coverage of other indexes). Additionally, it includes 100,000 institutions and all open-source codes and open APIs. To establish this as a viable and widely recognized open metrics and citations index, it is imperative that it be endorsed and recognized by UNESCO and relevant stakeholders to enhance its credibility, global recognition, and adoption.
- **Establishing a Global, More Fair, and Inclusive University and Institution Ranking:** To achieve this, the open institutions ranking needs to be based on the more inclusive open metrics platform, which ensures the widespread adoption of open metrics. The World Open University and Institution Rank should be established and recognized by UNESCO, towards democratizing the institution & academic ranking and valuation. This institution ranking index will help mitigate the negative impacts of biased and exclusionary existing institution ranking systems, that favor certain languages over others, thereby subjecting subject-fields, institutions, and countries to unequal treatment.
- **Providing substantial financial, technological, and training support** for the development of robust open science platforms in the Global South, aiming to alleviate the burden of APCs.
- **Actively supporting regional open science platforms**, by providing funding support, technical assistance, capacity building, and facilitating access to resources and technology, to promote collaboration and knowledge sharing within the Global South.
- **Facilitating access to AI capabilities** for developing countries, ensuring they can participate in and benefit from the advancements in this field, maybe through a call for true Open AI recommendation led by UNESCO.

Only through such proactive and sustained engagement can UNESCO ensure that the promise of open science is realized for all, fostering a truly inclusive and equitable global scientific community.